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What Do Visual Neural Networks Learn?

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Abstract

Over the past decade, artificial neural networks trained to classify images downloaded from the internet have achieved astounding, almost superhuman performance and have been suggested as possible models for human vision. In this article, we review experimental evidence from multiple studies elucidating the classification strategy learned by successful visual neural networks (VNNs) and how this strategy may be related to human vision as well as previous approaches to computer vision. The studies we review evaluate the performance of VNNs on carefully designed tasks that are meant to tease out the cues they use. The use of this method shows that VNNs are often fooled by image changes to which human object recognition is largely invariant (e.g., the change of a few pixels in the image or a change of the background or illumination), and, conversely, that the networks can be invariant to very large image manipulations that disrupt human performance (e.g., randomly permuting the patches of an image). Taken together, the evidence suggests that these networks have learned relatively low-level cues that are extremely effective at classifying internet images but are ineffective at classifying many other images that humans can classify effortlessly.

1. INTRODUCTION

Two decades ago, computer vision researchers often explained the difficulty of their study topic by contrasting it with the game of chess. While the best computer chess program can beat a human chess grandmaster, they explained, the most sophisticated computer vision systems still fall short of the visual performance of a 3-year-old child (e.g., Ullman 2004). Today, the first part of this statement is still true, but the second part no longer holds. Modern computer vision methods can now perform at an astonishingly high level, often surpassing well-trained humans in certain tasks. This dramatic increase in performance is largely due to a paradigm shift in the design of computer vision systems. Whereas previous systems relied to a large extent on understanding the problem to be solved and using this understanding to engineer solutions to the problem, modern methods are almost exclusively based on deep neural networks that learn how to perform vision based on examples. This naturally leads to the question: What do these successful neural networks learn?

In this article, we summarize experimental evidence from multiple studies addressing the classification strategy learned by successful neural networks. We focus on particular instances of visual neural networks (VNNs) that are often used in computer vision applications and in comparisons with human vision (e.g., Schrimpf et al. 2018, Sorscher et al. 2022). We start by detailing the architectures and learning strategies of these particular networks. Next, we give examples of the success of these models in a range of computer vision tasks and then discuss significant differences between how humans and VNNs classify images. Specifically, we discuss two kinds of differences: (*a*) excessive sensitivity, which causes VNNs to be easily fooled by image changes to which human object recognition is largely invariant, and (*b*) excessive invariance, which causes VNNs to yield similar responses to original images and to images that have undergone very large image manipulations that would disrupt human performance (e.g., randomly permuting the patches of an image). Finally, we discuss how VNNs relate to other approaches proposed throughout the history of computer vision and speculate on the performance that VNNs can reach in the future.

2. A BRIEF HISTORY OF COMPUTER VISION

In 1966, Seymour Papert, an MIT professor working on artificial intelligence, assigned a group of students the following summer project:

> The primary goal of the project is to construct a system of programs which will divide a vidisector picture into regions such as likely objects, likely background areas and chaos. We shall call this part of its operation FIGURE-GROUND analysis. It will be impossible to do this without considerable analysis of shape and surface properties, so FIGURE-GROUND analysis is really inseparable in practice from the second goal which is REGION DESCRIPTION. The final goal is OBJECT IDENTIFICATION which will actually name objects by matching them with a vocabulary of known objects. (Papert 1966)

The fact that Papert assigned such an ambitious project to his summer students is indicative of a recurring theme in the history of computer vision: the tendency to underestimate the difficulty of the problem. While vision seems effortless, computer vision researchers have found it notoriously difficult to replicate human performance. Thus, 20 years after Papert, Berthold Horn, also a professor at MIT, wrote in the introduction of his book *Robot Vision* that "vision is our most complicated sense" and "it is not surprising, then, that many attempts to provide machines with a sense of vision have ended in failure" (Horn 1986, pp. 1–2).

Horn's book largely follows the outline of Papert's project: It describes methods to segment images into coherent regions, then discusses methods to describe the properties of surfaces, and ends by discussing methods to recognize objects and 3D relationships between objects. The main thesis of the research summarized in Horn's book is that "machine vision should be based on

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a thorough understanding of image formation. This emphasis allows us to derive mathematical models of the image-analysis process. Algorithms for recovering a description of the imaged world can then be based on these mathematical models" (Horn 1986, p. 13).

The philosophy and methods used in Horn's 1986 book formed the backbone of much of computer vision for the next 25 years. Thus, Richard Szeliski's book from 2010 (Szeliski 2010) follows a structure similar to Horn's and similarly argues for understanding image formation and for mathematical formulations. The book is striking for the wide range of methods and approaches to computer vision it covers: an engineering approach (i.e., careful specification of the problem, the constraints, and possible solutions), a scientific approach (i.e., understanding the physics of image formation), a statistical approach (i.e., modeling uncertainty in images and the possible representations), and an algorithmic approach (i.e., efficient and robust algorithms for computer vision). Szeliski summarizes the state of affairs in 2010 as follows: "Despite all of these advances, the dream of having a computer interpret an image at the same level as a two-year old (for example, counting all of the animals in a picture) remains elusive" (Szeliski 2010, p. 4).

In the summer of 2012, the results of the ImageNet Large Scale Visual Recognition Challenge were announced. The challenge measured the success of computer vision systems in assigning 1 of 1,000 possible textual labels (e.g., "dalmatian dog") to an image (Russakovsky et al. 2015). By far, the best-performing method was a VNN called AlexNet (Krizhevsky et al. 2012). AlexNet differed from the classical recognition methods in computer vision in two fundamental ways. First, unlike the process suggested by Papert, in which images are first segmented into regions, then the surface properties of these regions are estimated, leading to the recognition of objects, AlexNet simply treated the recognition process as one black box that maps images to textual labels. Second, unlike the approach advocated by Horn, Szeliski, and others, AlexNet was not based on understanding image formation and a mathematical formulation of the vision problem. Rather, the system learned by itself how to perform recognition from a large set of labeled examples.

The success of AlexNet had a dramatic effect on all of computer vision. This effect can be summarized by a quote from an article in *The New Yorker* (Seabrook 2020):

> Rama Chellappa, a professor at the University of Maryland who is one of the top researchers in the field, told me, "Let me give you an analogy. Let's assume there are ten major religions in the world. What if, after 2012, everything became one religion? How would that be?" With computer-vision methods, he said, "that's where we are."

Indeed, the approach of solving vision tasks with VNNs that learn how to solve the tasks by themselves from examples now dominates the field of computer vision, enabling a tremendous increase in performance. As a result, problems that were considered intractable for many decades are now considered essentially solved. In particular, the problem of counting animals in photographs, referred to as elusive by Szeliski (2010), is now performed by VNNs with the same accuracy as crowdsourced human volunteers (Norouzzadeh et al. 2018). Our goal in this article is to describe what such successful VNNs have learned.

3. SPECIFIC INSTANCES OF VISUAL NEURAL NETWORKS

Given that almost any paper published these days in the premier computer vision conference proceedings uses VNNs and that there are thousands of such papers every year, it is impossible to answer the question of what all modern VNNs have learned. In this article, we focus on eight particular VNNs. We choose them because they are repeatedly used as benchmarks for image classification and are also as comparisons in a number of human vision studies (e.g., Schrimpf et al. 2018, Sorscher et al. 2022). The eight VNNs are listed in **Table 1**. They represent a change over time in both the architectures and the training procedures of VNNs.

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Table 1 The eight visual neural networks focused on in this article

	Dataset (size)	Top-1 accuracy	Number of parameters	Reference
AlexNet	ImageNet (1.28 M)	62.5%	60 M	Krizhevsky et al. 2012
VGG	ImageNet (1.28 M)	74.5%	144 M	Simonyan & Zisserman 2015
ResNet50	ImageNet (1.28 M)	76.5%	25.6 M	He et al. 2016
ViT-B/16	JFT-300M (300 M)	84.15%	86.6 M	Dosovitskiy et al. 2021
ViT-H/14	JFT-300M (300 M)	88.04%	632 M	Dosovitskiy et al. 2021
CLIP (ViT-L/14)	WIT (400 M)	76.2%	370 M	Radford et al. 2021
OpenCLIP (ViT-G/14)	LAION-2B (2 B)	80.1%	2.5 B	Ilharco et al. 2021
DINOv2 (ViT-G/14)	LVD-142M (142 M)	86.5%	1.1 B	Oquab et al. 2024

Top-1 accuracy refers to the ImageNet task in which the visual neural network is asked to assign 1 out of 1,000 possible labels to the image (so that chance performance is 1/1,000).

3.1. Architectures

When VNNs learn to solve a visual task using examples, they do so by updating a large set of parameters in order to improve performance on the particular task. However, certain aspects of the VNNs remain fixed throughout this process. These aspects are referred to as the architecture of the network. In particular, the architecture determines the number and types of layers that are present in the VNN, as well as the functional form of the interaction between different units.

Going back to AlexNet, the first generation of VNNs used an architecture called a convolutional neural network (CNN), but over the last decade, most successful VNNs have begun to use an alternative architecture called a vision transformer (ViT). We briefly explain the differences between these two architectures.

As the name suggests, each layer of a CNN performs a convolution of the representation in the previous layer, and the convolution is followed by a nonlinearity. Thus, the activity of a neuron in a given layer is a weighted sum of the activities of neighboring neurons in previous layers, and the weights used in the sum are the same at all locations. The filter used in the convolution is relatively small (a typical size is 3×3 pixels for 256×256 images), which means that the receptive field of neurons gradually grows as we go deeper into the network: The first layers have small receptive fields and neurons are sensitive to small parts of an image (e.g., an edge), while the deeper layers have larger receptive fields and neurons respond to objects (or parts of objects) (Bolei et al. 2015). This has naturally led to comparisons with biological vision systems where receptive fields also grow gradually (e.g., Schrimpf et al. 2018, Sorscher et al. 2022). Krizhevsky et al. (2012) justified the use of convolutions by the need to incorporate prior knowledge into the learning process. They argued that CNNs make "strong and mostly correct assumptions about the nature of images (namely, stationarity of statistics and locality of pixel dependencies)" (Krizhevsky et al. 2012, p. 1097). The first three VNNs that appear in **Table 1**—AlexNet, VGG, and ResNet—are all successful examples of CNNs, supporting the claim that the assumptions they make about the nature of images are mostly correct.

Over the last 5 years, CNNs have gradually lost their dominance among VNNs and are being replaced by a newer architecture, ViT. This architecture is based on the success of transformers in natural language processing (Vaswani et al. 2017): The T in ChatGPT stands for transformer. Among the VNNs described in **Table 1**, ViT, CLIP, and DINO are successful examples of ViTs. In a ViT, images are first converted into tokens, which are the analogs of words in natural language processing. This conversion is done by dividing the image into nonoverlapping patches (typically 16×16 patches for 256×256 images), then each layer nonlinearly transforms the representation

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of all tokens in the image. Importantly, the nonlinear transformation is the same at all image locations but is nonlocal: At each layer, the representation of a particular token is a weighted average of the representations at all other tokens in the image. The weights that a token assigns to other tokens in the image are referred to as the self-attention map, which defines the receptive field of a neuron in an image-dependent manner. In particular, even in the very first layer, the receptive field may be very large (Raghu et al. 2021). Thus, a ViT makes strong assumptions about images and the stationarity of their statistics, but it allows for much more global interactions. This architecture also seems further removed from models of biological visual systems.

The use of either CNNs or ViTs reduces the number of parameters in a VNN compared to fully connected architectures (in which all neurons in one layer are connected to all neurons in the subsequent layer), but the number of parameters is still tremendously high. As shown in **Table 1**, the number of parameters ranges from 25 million to more than 2 billion.

3.2. Training Methods

In addition to the architecture of a VNN, training a VNN requires specifying two additional aspects that control what the VNN will learn: (*a*) the training set and (*b*) the loss function. The training set used to train the first generation of successful VNNs is ImageNet (Deng et al. 2009), a dataset of roughly 1 million images annotated by humans. The annotation consists of a single category that describes the full image (e.g., "dalmatian dog," "photocopier"). The loss used in this early generation of models is known as a supervised loss or end-to-end training. This loss completely ignores all internal representations computed by the network and focuses on only the final label the network assigns to a given image. It compares this label to the category supplied by a human annotator on the same image and adjusts the parameters to make the predicted category more accurate. This paradigm of training mostly on ImageNet using purely supervised learning was used to train the first three models in **Table 1**: AlexNet, VGG, and ResNet. However, the more modern VNNs listed in the table use much larger datasets and possibly additional training losses.

Whereas ImageNet contains approximately 1 million images, ViT, CLIP, and DINO were trained on datasets with hundreds of millions of images. ViT was trained on the JFT-300M dataset, which contains 300 million images with automatic labels in 18,291 categories (Sun et al. 2017). The labels are obtained using "an algorithm that uses complex mixture of raw web signals, connections between web-pages and user feedback" (Sun et al. 2017, p. 845). Once these labels are obtained, the training loss is similar to the one used to train CNNs: Parameters are adjusted so that the category predicted by the network will agree with the given labels of the training images.

Starting with CLIP, models have increasingly been trained with natural language supervision, rather than an individual word per image. Furthermore, to scale the number of training images, the models are based on publicly available image sources. CLIP used a proprietary dataset of 400 million images "collected from a variety of publicly available sources on the Internet" (Radford et al. 2021, p. 8749). OpenCLIP, as the name suggests, created an open dataset that is available to the broader research community (Schuhmann et al. 2022, Cherti et al. 2023) by leveraging the accessibility text that commonly appears in web pages that contain images: When a web designer adds an image to a web page, she will often include a short sentence describing the image for the benefit of visually impaired users. As an example, the following accessibility text was used to describe the image headlining the home page of the *New York Times* website on February 23, 2025: "Jenny Boyter poses behind a small bar with a case of Tab, holding a can in her left hand" (Kurutz 2025). By leveraging these rich, textual descriptions, OpenCLIP created a dataset of over 5 billion images that can be used to train VNNs without the need for additional

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human annotations. In both CLIP and OpenCLIP, the combination of text and images was used to learn the parameters of two separate deep neural networks: a visual network that converts the image into a vector of numbers (known as visual embedding) and a text network that converts the textual description into a second embedding vector (known as text embedding). The parameters of the model are adjusted to make the embedding of the text align with the embedding of the image.

DINO was trained using a method called self-supervised training. Given a training image, two views of that image are generated (e.g., by randomly cropping two subimages from the image or randomly changing the color map by a small amount). In self-supervised training, the parameters of the neural network are adjusted so that the two views of the same image will have a similar output while also avoiding trivial solutions. This method requires no labeled data and the best-performing version of DINO, DINOv2, was trained on 140 million unlabeled images (Oquab et al. 2024).

Even when billions of images are available for training, it is common practice to add additional training examples using data augmentation. Each training image is transformed using a transformation that is assumed to preserve the label of the image. AlexNet was trained using two types of transformations: image translations and reflections. This is done by (*a*) choosing a random 224×224 subimage from a 256×256 image and possibly reflecting the image horizontally, thus effectively increasing the dataset size by a factor of 2,048, and (*b*) modifying the RGB values of the image by adding a single randomly selected vector to all pixels. All modern training methods include these two augmentations along with additional ones (e.g., scaling the image by a random amount).

To summarize, existing VNNs can vary greatly in their architectures, training procedures, and training sets. At the same time, there exist two significant commonalities between all successful VNNs:

1. Successful VNNs have a tremendously high number of parameters, and these parameters are set automatically by a learning procedure that operates on a training set.
2. Successful VNNs are almost exclusively trained on images downloaded from the internet.

Given these commonalities, we expect that all successful VNNs will learn a somewhat similar classification strategy. As we show in the next section, this is indeed the case.

4. SUCCESS STORIES

As mentioned in the introduction, the deep learning revolution in computer vision started with the success of AlexNet in the 2012 ImageNet Large Scale Visual Recognition Challenge. Success in this challenge was measured by asking whether the category predicted by the network for a new image (i.e., an image not seen during training) was the correct one (known as top-1 accuracy), or whether one of the five categories predicted by the network was the correct one (known as top-5 accuracy). The top-5 accuracy of ResNet is around 97%, and the top-5 accuracy of naive humans is estimated to be less than 95% (Russakovsky et al. 2015). The top-1 accuracies of the methods discussed in this article are shown in **Table 1**. This accuracy is continuously improving, and for methods such as CLIP and DINO, it is close to 90% (chance performance is less than 1/1,000). This is astonishing performance and is better than human performance on the same task (He et al. 2015).

Over the last decade, however, the vision community has grown increasingly critical of the use of ImageNet accuracy as a measure of VNN performance. First, many images in the ImageNet dataset contain several equally prominent objects, and the choice of one of them as the true label is arbitrary. Second, the labeling of the annotators is based on a predefined set of labels that

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sometimes seems haphazard: For example, the dataset contains separate labels for "sunglass" and "sunglasses" and also distinguishes between "laptop" and "notebook" computers. In this sense, some of the astonishingly high top-1 accuracies that VNNs achieve on ImageNet may be "due to overfitting to the idiosyncrasies of their labeling pipeline" (Beyer et al. 2020, p. 2).

A second way to measure VNN performance is to use the trained networks as a backbone with which different visual tasks can be solved. For example, given a neural network such as VGG, which was trained to perform object categorization, one can train a network to use the internal representation of VGG to perform semantic segmentation (i.e., label all pixels in which a particular object appears). It has repeatedly been found that when using the VNNs listed in **Table 1**, it is possible to solve a wide range of downstream tasks that all use the same learned representation.

Figure 1 shows examples of downstream tasks that have been successfully solved with the internal representations of VNNs trained to perform image classification. Semantic segmentation refers to the task in which each pixel in an image is labeled as belonging to one of a set of possible categories. In object detection, a bounding box is calculated for specific objects. In depth estimation, a depth map is calculated where the color of each pixel indicates its depth, and in sparse matching, a set of corresponding key points is matched between a pair of given images.

While the standard task of image classification requires a computer to determine what is in the image, the other tasks shown in **Figure 1** require a computer to determine where different components in the scene are located. A long-standing hypothesis in neuroscience is that different pathways are devoted to these two types of tasks (de Haan & Cowey 2011). In contrast, VNNs can achieve state-of-the-art performance in both types of tasks by using the same representation. In particular, training a linear readout layer on top of frozen DINO features has been shown to yield

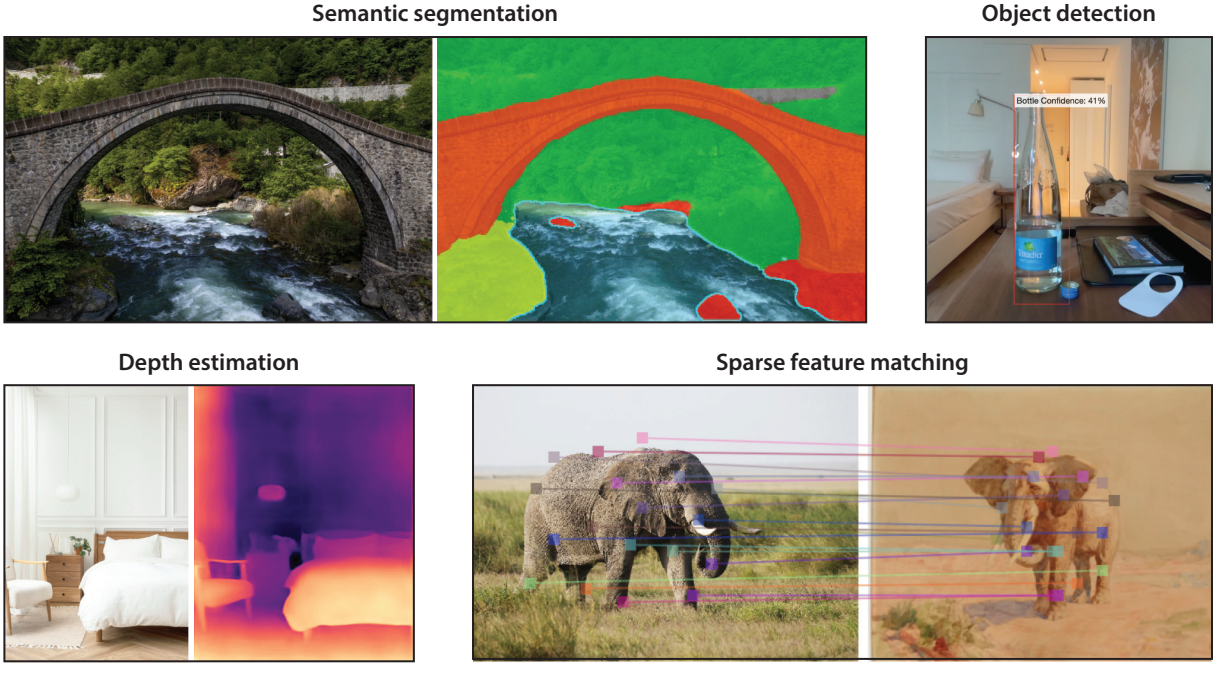


Figure 1

Visual neural networks achieve astounding performance on a range of different tasks. The semantic segmentation, depth estimation, and sparse feature matching results were obtained by running the DINOv2 algorithm (Oquab et al. 2024). The object detection results were obtained by running the YOLO algorithm (Redmon et al. 2016).

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excellent performance in image categorization (a what task), as well as depth estimation, semantic segmentation, and sparse feature matching (where tasks) Oquab et al. (2024).

5. APPROACHES TO THE UNDERSTANDING OF HOW VISUAL NEURAL NETWORKS WORK

How do VNNs achieve this excellent performance? Since they have millions of parameters that are estimated using a training procedure, there is no easy way to measure what they have learned or characterize the cues they use to solve certain tasks.

One class of approaches attempts to identify single neurons in the VNNs that correspond to particular concepts. Early approaches were mostly based on visualization of the pixels or image regions that are most likely to influence the output of a particular neuron (e.g., Zeiler & Fergus 2014, Sundararajan et al. 2017). More modern methods use semiautomatic tools (Bau et al. 2020) to comprehensively analyze many neurons in the network. The most recent approaches use large language models to automatically analyze and describe the functions of subsets of neurons in a VNN (Shaham et al. 2024).

In this article, we review papers that take an alternative approach. This approach attempts to understand what VNNs have learned by evaluating their performance on carefully designed tasks that are meant to tease out the cues they use. This is similar to the approach of investigating human vision by studying the mistakes people make on carefully designed stimuli, as is done in the study of visual illusions. We focus primarily on the image classification task, where VNNs are asked to output a text label that describes what is in the image. The papers we review suggest that the strategy used by VNNs to solve this task seems to be very different from that used by humans. We divide the papers we survey into two groups: (*a*) those that show an excessive sensitivity of the VNNs to changes that do not influence human categorization and (*b*) those that show an excessive invariance of the VNNs to changes that completely disrupt human categorization.

5.1. Excessive Sensitivity

One year after the publication of the AlexNet paper (Krizhevsky et al. 2012), it was pointed out that VNNs that perform image classification at such a high degree of accuracy are highly sensitive to small, imperceptible perturbations of their input image (Bruna et al. 2014, Goodfellow et al. 2015). **Figure 2*a*** shows an example: By adding a small amount of carefully designed noise to the input image, one can fool a state-of-the art VNN and make it classify an image of a cat as guacamole with perfect confidence (Tramèr 2023). The images created by adding this carefully designed noise are now called adversarial images, and it is possible to fool VNNs by causing the network to make any incorrect prediction (i.e., untargeted attack) or by forcing the network to classify the input as a specific class (i.e., targeted attack). Thus, another noisy perturbation of small magnitude can make the same image of a cat be classified by the VNN as a goldfish, a revolver, or any other target label (Hendrycks 2023). This is a notoriously infamous vulnerability of neural networks, and, therefore, there are countless works on the topic (e.g., Goodfellow et al. 2015, Moosavi-Dezfooli et al. 2017, Athalye et al. 2018, Kurakin et al. 2018, Xiao et al. 2018).

How are these adversarial examples found? The most common method is to compute the gradient of the VNN's output with respect to the input pixels and perturb the input image by a small amount in the direction of the gradient. This is known as a white box attack because the adversary, who is searching for the perturbation, has access to the VNN and can compute the gradient. There are also black box attacks, where the adversary does not have access to the particular VNN. For example, universal perturbations (Moosavi-Dezfooli et al. 2017) are computed by having access to one VNN and a set of training images. The same imperceptible perturbation is then applied to other VNNs as well as other images and often succeeds in changing the new VNN's output on

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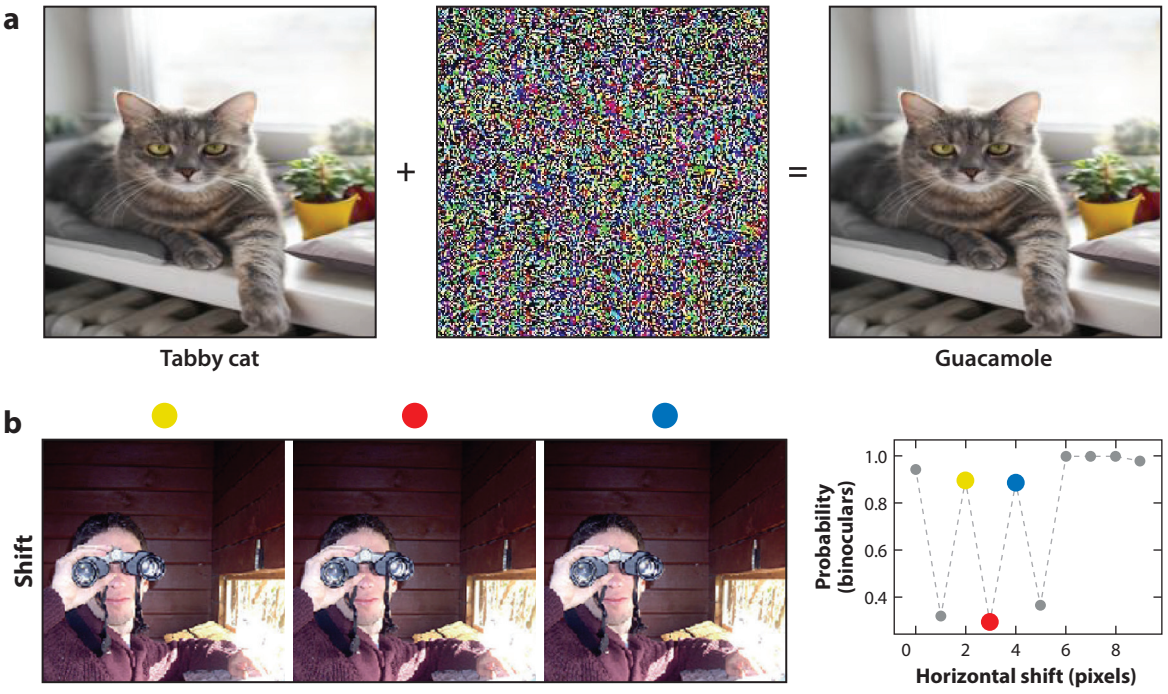


Figure 2

(*a*) Adversarial attacks demonstrating the excessive sensitivity of the models to tiny perturbations. By adding a small amount of noise, an adversary can cause a state-of-the-art visual neural network to classify an image of a cat as guacamole. (*b*) A similar effect with very small pixel space shifts that the human eye can barely perceive. The graphs show the probability of the correct label being computed by the visual neural network (VNN) for the shifted images (marked in *yellow*, *red*, and *blue*). Panel *b* adapted with permission from Azulay & Weiss (2019).

the new images. Perhaps the simplest method for finding a noise pattern that will fool a particular VNN is to repeatedly add random noise to the input until the network's output is changed. Depending on the amount of noise that is added, as few as 100 attempts may be sufficient to find at least one noise pattern that fools the VNN (Webb et al. 2019).

The standard way to measure the sensitivity of VNNs to adversarial examples is by computing a quantity called the adversarial accuracy of the model. To understand how adversarial accuracy is computed, we can contrast it with the standard notion of accuracy, known as clean accuracy. The clean accuracy of a VNN is shown in the rightmost column of **Table 2**. It is measured as the fraction of times a VNN correctly classifies an image, and as we discussed in the previous section, for modern VNNs, this number (i.e., top-1 accuracy) is over 80%. The adversarial accuracy, on the other hand, is the fraction of images a VNN correctly classifies even after an adversary is allowed to change each pixel by a small amount (typically 16/256 for ImageNet images). A recent study (Mahmood et al. 2021) found that the adversarial accuracy of most VNNs is 0% [i.e., for 100% of the images, it is possible to fool state-of-the-art VNNs with an imperceptible perturbation in which each pixel is changed by less than 16/256 (see **Table 2**)]. This excessive sensitivity is true for all of the networks we have discussed, including ResNet and ViTs. In particular, it is true for ViTs trained on hundreds of millions of examples.

A possible criticism of standard adversarial examples is that they rely on unrealistic noise patterns, but it is also possible to fool deep neural networks with imperceptible, realistic changes to

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Table 2 Quantification of the prevalence of adversarial examples using several different white box attacks (Mahmood et al. 2021)

	FGSM	PGD	BPDA	MIM	C&W	APGD	Top-1 accuracy
ViT-B-16	23.1%	0.0%	7.3%	0.0%	0.0%	0.0%	80.3%
ViT-L-16 (224)	27.9%	0.0%	8.4%	0.0%	0.0%	0.0%	82.0%
ViT-L-16 (512)	29.8%	0.0%	8.4%	0.0%	0.0%	0.0%	85.4%
BiT-M-R50x1	28.7%	0.0%	3.5%	0.0%	0.0%	0.0%	79.9%
BiT-M-R152x4	60.9%	0.0%	15.2%	0.0%	0.0%	0.0%	85.3%
ResNet-50	11.8%	0.0%	1.4%	0.0%	0.0%	0.0%	74.5%
ResNet-152	18.1%	0.0%	2.7%	0.0%	0.0%	0.0%	77.0%

Most attacks [PGD (Madry et al. 2018), MIM (Dong et al. 2018), C&W (Carlini & Wagner 2017), and APGD (Croce & Hein 2020)] were able to find an adversarial example for all of the models tested and for 100% of the images in ImageNet. Abbreviations: APGD, auto projected gradient descent; BPDA, Backward Pass Differential Approximation; C&W, Carlini and Wagner; FGSM, fast gradient sign method; MIM, momentum iterative method; PGD, projected gradient descent.

the image. **Figure 2*b*** shows results from the work of Azulay & Weiss (2019): By translating the image by a few pixels, the classifications outputted by a VNN can vary greatly. In a recent paper, Shifman & Weiss (2024) showed that this excessive sensitivity to single-pixel shifts continues to hold for modern VNNs trained on hundreds of millions of examples (e.g., CLIP, DINO).

To quantify the sensitivity to small translations of the image, Shifman & Weiss (2024) conducted experiments with image retrieval. They used the internal representation of VNNs as an image descriptor and used these descriptors to retrieve the most similar image to a query image (out of approximately 1,000 other images in the same category). They asked whether the retrieved images will be the same for a query image and all 1-pixel shifts of the query image. As shown in **Table 3**, for more than 40% of the images, there exists a simple 1-pixel shift of the query image that will change the retrieved image. Thus, while a single-pixel shift is almost imperceptible to a human observer even with prolonged viewing (compare, for example, the three images in **Figure 2*b***), the top-performing VNNs are highly sensitive to these shifts. Another example is change of viewpoint: A recent study showed that for the majority of ImageNet images, it is possible to change the output of CLIP and ResNet simply by shifting the viewpoint of the camera by a tiny amount (Madan et al. 2025).

Table 3 Quantification of the sensitivity of visual neural networks to realistic shifts (Shifman & Weiss 2024)

Model	Accuracy	Adversarial accuracy shift size			
		1	3	5	9
Vanilla ViT	76.25%*	37.21	14.52	8.93	6.40
ResNet 50	76.74%	46.33	25.08	17.44	10.39
ConvNext-T	78.29%	59.16	37.53	29.07	21.20
ConvNext-L	83.47%	65.58	45.70	37.35	29.13
OpenCLIP	69.33%*	57.01	35.28	27.81	22.22
DINOv2	84.11%	62.42	40.74	32.68	26.91

For DINOv2, for example, for 40% of the images, the output can be changed when the input image is shifted by just 1 pixel. This increases to more than 70% when 9-pixel shifts are used. An asterisk indicates that the accuracy was evaluated using a k-nearest neighbors classifier on the model's representation.

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A second class of papers (Beery et al. 2018, Alcorn et al. 2019, Barbu et al. 2019, Geirhos et al. 2019, Bordes et al. 2024) demonstrates excessive sensitivity to perturbations that are large enough to be clearly noticed by humans but are ignored by humans when deciding the category of the image (**Figure 3**). For example, when we take a picture of a cow in a grassy field and replace the background with that of a beach, humans will notice that a change has been made, but the cow will remain a cow. In contrast, it has been repeatedly shown that state-of-the-art VNNs can change their categorization of the object when the background is changed (Beery et al. 2018, Barbu et al. 2019, Bordes et al. 2024).



Figure 3

Examples of perturbations that are large enough to be clearly noticed by humans but are ignored by humans when deciding the category. (*a*) Output of a commercial image labeler on real images of cows in different backgrounds. (*b*) Examples of computer graphics images from the Photorealistic Unreal Graphics (PUG) dataset along with the output of ResNet50 and a vision transformer (ViT). When the background or texture is not common, visual neural networks often fail to correctly classify the animal. These effects are quantified in **Table 4**. Panel *a* adapted with permission from Beery et al. (2018). Images in panel *b* reproduced with permission from Bordes et al. (2024).

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Another example is texture: Geirhos et al. (2019) asked how changing the texture of the object might affect the resulting output of both humans and models. They took images from different classes of ImageNet and changed only the texture of the objects (images were chosen so that the original images were correctly recognized by the VNNs and humans). The results were definitive: While humans were able to ignore the change and still recognize the original object, the VNNs showed a clear bias toward the texture (i.e., classifying the images by the object the texture belonged to).

In an attempt to quantitatively measure the sensitivity of modern VNNs to changes in parameters such as background and texture, Bordes et al. (2024) constructed various computer graphics–generated datasets for which different factors of variation were controlled for (background, texture, size, lighting, viewpoint, etc.). Using these datasets, they could measure the robustness of the models to different changes. For example, they took a picture of an elephant and changed the color of the elephant's skin to white. Of course, humans notice the change but still see an elephant. In contrast, state-of-the-art VNNs are often fooled by such a transformation and label the elephant as an Ibizan hound (**Figure 3*a***). A numerical quantification of the results is given in **Table 4**. The accuracy of the models drastically drops from a clean accuracy of around 80% on ImageNet to less than 30% when the background changes and 15% when the lighting changes. As can be seen in the leftmost column of **Table 4**, this excessive sensitivity holds for models trained on hundreds of millions of images or even billions of images. A similar drop in performance was also reported by Barbu et al. (2019) for real images in which objects were photographed with different backgrounds and viewpoints.

5.2. Excessive Invariance

While the previous section detailed experiments showing how the output of a network can be changed by an imperceptible change to the image, a second stream of research shows the complementary effect. That is, the output of a network can be completely stable even under extremely large perturbations that totally disrupt human perception.

One example of this excessive invariance is model metamers (Feather et al. 2023), which are pairs of images that elicit exactly the same response from a network even though they are designed to be perceptually distinct. In other words, for any given image, a metamer is another image that is very different visually but yields exactly the same activation pattern for a chosen layer of interest. **Figure 4*a*** shows an example of such a pair. The metamer image was constructed by starting from a random image and gradually modifying the image so that its representation approaches that of the

Table 4 Performance of visual neural networks on synthetically generated images where factors of variation can be controlled (Bordes et al. 2024)

Model	ImageNet top-1 accuracy	Size	Texture	Lighting	Background
ResNet50	81.5	35.7	27.0	13.6	29.5
ResNet101	82.3	39.7	31.1	14.1	32.8
ViTLarge	85.8	46.4	42.9	8.9	34.6
ViTBasePretrained21k	84.3	33.0	28.5	4.1	26.6
DINOv2 (LVD-142M)	84.5	43.1	35.0	6.1	30.9
CLIPVITB32 (400 M)	62.9	40.1	34.4	5.7	24.4
CLIPVITB32 (2 B)	66.6	42.2	34.7	3.3	26.0
CLIPVITL14 (400 M)	72.8	50.6	41.1	4.3	33.0

The accuracy of the models drop drastically from a clean accuracy of around 80% on ImageNet to less than 30% when the background changes and 15% when the lighting changes.

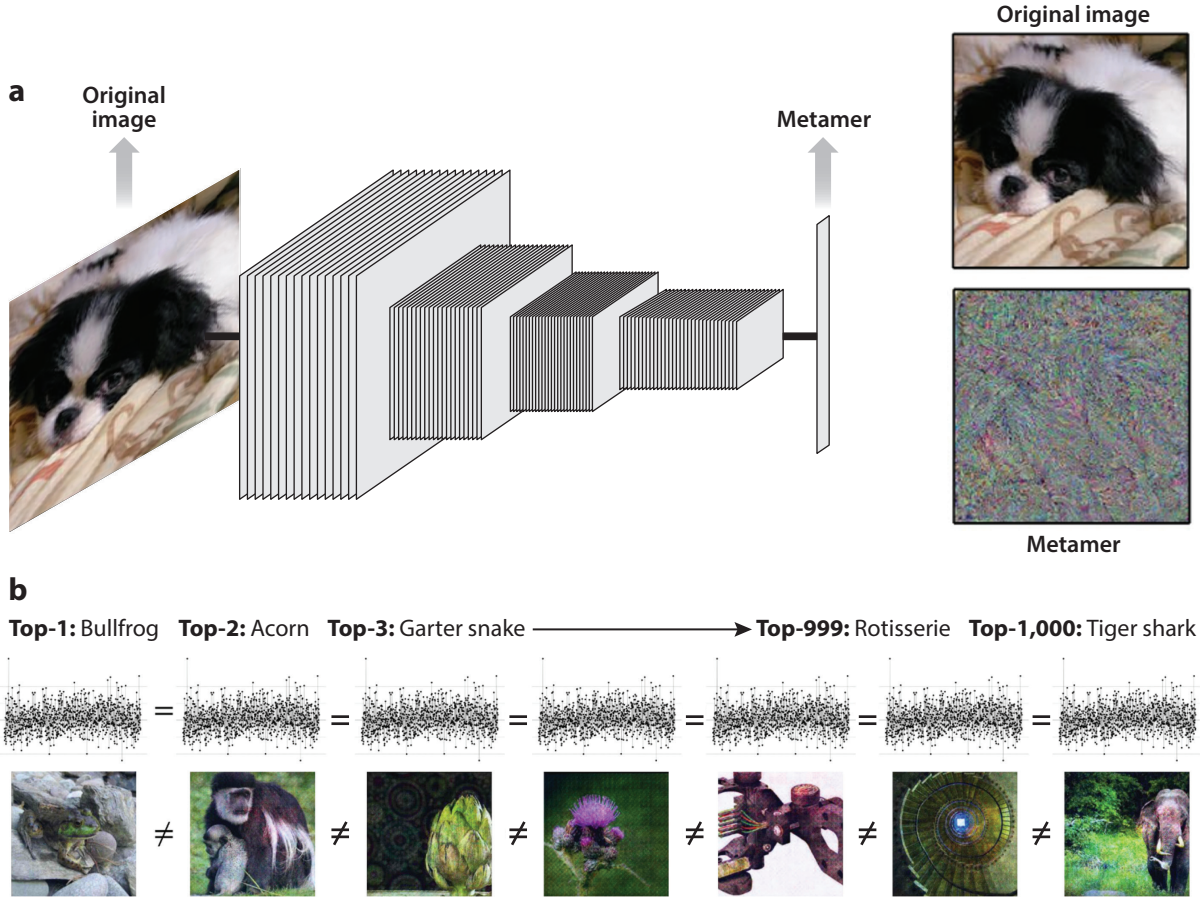


Figure 4

Metamers are pairs of images that elicit exactly the same response from a network. (*a*) The noise image shown in the top right elicits exactly the same response in ResNet50 as the image of the dog shown below it. Similar results were reported for additional visual neural networks (VNNs), including CLIP and vision transformers. (*b*) A series of images that all yield the exact same activation pattern in the final layer of a VNN, even though they are clearly images of different objects. Panel *a* adapted with permission from Feather et al. (2023). Panel *b* adapted with permission from Jacobsen et al. (2019).

target image. While no human would describe the metamer image as a dog, it elicits precisely the same response from the network as the original dog image shown above it. The existence of such metamers reveals a major difference in the generalization of models and humans, emphasizing the divergent invariances the two exhibit.

Another algorithm for constructing metamers was presented by Jacobsen et al. (2019). They searched for images that were close to an image from a different class but still yielded the same response in the final layer of a VNN as the target image. As shown in **Figure 4*b***, all of the images on the bottom elicit exactly the same response in the final layer of a VNN, yet they are clearly images of different objects. Again, this emphasizes the excessive invariance of modern VNNs.

Metamers are optimized only for eliciting identical activations. They are not restricted to maintaining any relations to their originating seeds. To better understand the invariances, other researchers use a more structured approach, that is, interpretable transformations to images that do not affect the model's output. **Figure 5*a*** demonstrates such a case. We see the effect of applying

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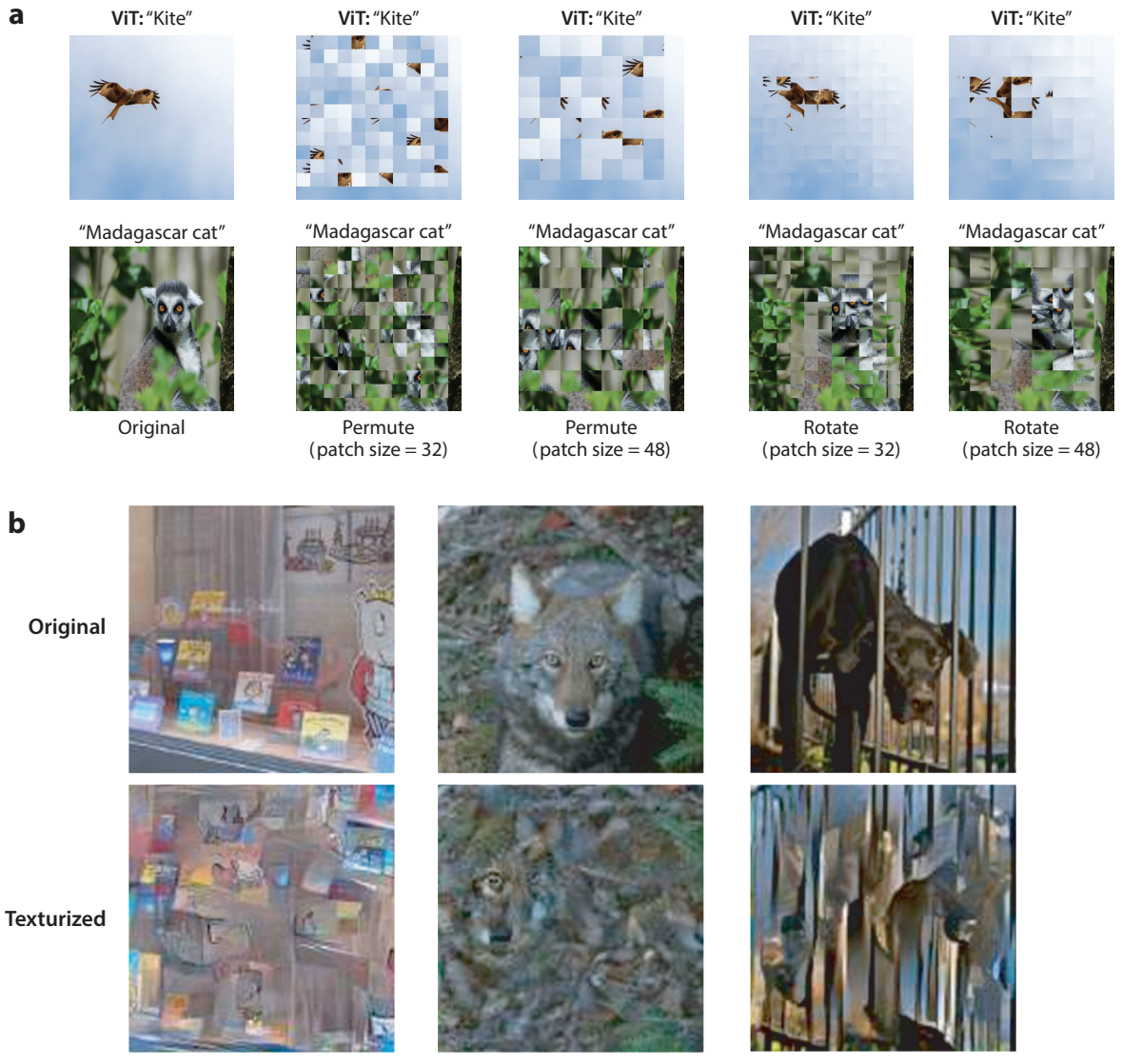


Figure 5

(*a*) Different patch transformations (permuting patches or rotating patches) hardly change the output of the vision transformer (ViT). While the images demonstrate how these transformations completely destroy the original image to the human eye, Qin et al. (2022) have shown that ViTs can classify such permuted images at an astonishing level (ImageNet top-1 accuracy above 60%). (*b*) The images are constructed so that their texture descriptors are similar to those of the original image, but they lack any global structure. While these transformations completely disrupt human perception, some high-performance visual neural networks are almost invariant to them. Panel *b* adapted with permission from Brendel & Bethge (2019).

different patch operations to input images. These alterations have very little effect on the network's output (Qin et al. 2022), even though randomly permuting the patches in an image destroys all of the shape information and makes it difficult to perceive the object for humans. State-of-the-art ViTs are almost equally good at classifying these permuted images as they are with the original image. A similar behavior can be seen when rotating patches or replacing them with background

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patches. These results emphasize another distinct property of VNNs: their heavy reliance on local, low-level cues that allow them to continue to perform well even when all global shape information has been removed from the images.

Another transformation that makes it impossible to perceive the shape of objects is texturizing (Brendel & Bethge 2019). As shown in **Figure 5*b***, this process is similar to permuting the location of patches in the image, but it avoids patch artifacts: The images are constructed so that their local statistics are similar to those of the original image, but they lack any global structure. Brendel & Bethge (2019, p. 5) showed that VGG "still reaches high accuracy on the texturised images while humans suffer greatly from the loss of global shapes in many images." This is another example of how successful VNNs are almost invariant to transformations that disrupt human perception.

5.3. Excessive Sensitivity and Invariance Beyond Image Categorization

While the preceding section focused on image categorization, excessive sensitivity and invariance of VNNs has also been reported in very different tasks. We give some examples below.

Carefully designed adversarial examples can greatly disrupt the performance of state-of-the-art VNNs on depth estimation and object detection. This is true even for realistic scenarios such as self-driving cars. A recent study found that by physically placing a particular pattern on the rear of a vehicle, one can "cause over 6 meters mean depth estimation error and reduce the object detection rate from 90.70% to 5.16%" (Cheng et al. 2022, p. 514). Note that no manipulation is done to the image pixels in this case; rather, a physical object is placed on the car and causes the VNN to fail to detect the car when this car is imaged with a conventional camera.

The sensitivity to translations of a few pixels that we discussed in the context of image categorization has also been demonstrated in semantic segmentation. For example, a VNN that correctly labels all pixels in the rear view of a car as belonging to the class "car" may fail to correctly label almost 50% of the pixels when the input image is shifted by a few pixels (Saha & Gokhale 2025).

More generally, recall from our discussion in Section 4 that modern computer vision often uses the same frozen representation for a wide range of tasks, from what tasks, such as image classification, to where tasks, such as depth estimation and semantic segmentation. This means that almost all of the failures we have discussed in the context of image classification will automatically transfer to additional tasks. Specifically, consider the construction of image metamers, which we discussed in Section 5.2. These metamers elicit exactly the same internal representation in a given layer of a given VNN as a specific real image. Hence, the image category given to these metamers will be the same as that of the real image, and it will hold for a range of additional tasks that use the same representation. In particular, the noisy metamer image shown in **Figure 4** will give exactly the same depth estimate as the original image of the dog for a method that uses frozen ResNet features together with a readout layer for depth estimation.

6. DISCUSSION

The main question we asked in this article was: What classification strategy have VNNs learned that allows them to be so successful at classifying internet images? We reviewed two differences between how VNNs classify images and how humans perform the same task. First, VNNs are excessively sensitive to perturbations that do not disrupt human classification (i.e., single-pixel shifts). Second, VNNs are excessively invariant to perturbations that may completely disrupt human perception of the same images (e.g., patch permutation).

Taken together, the evidence suggests that these networks have learned relatively low-level cues that are extremely effective at classifying internet images but are ineffective at classifying many

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other images that humans can classify effortlessly. As a thought experiment, consider a classifier that can correctly classify an image as "cat" or "dog" by computing the ratio of brightnesses in the first pixel to that in the last pixel of the image. For the sake of argument, suppose that such a classifier is able to classify all images in the ImageNet test set perfectly. Even if such a classifier existed, it would be easily fooled by small, imperceptible changes to the image (e.g., changing just the first pixel). At the same time, such a classifier may be invariant to very large changes in the image that would completely disrupt human perception (e.g., changing all pixels except for the first and last ones). While we are by no means suggesting that the classification strategy learned by successful VNNs is as simplistic as the one in our thought experiment, the papers we reviewed suggest that the strategy is closer to that extreme than to a classical approach based on segmenting the object from the background and identifying the object's shape.

In this context, it is interesting to ask: Have VNNs accomplished the goals of the summer project Papert assigned to his students in 1960? On the one hand, current VNNs are astonishingly good at the last goal of Papert's project (i.e., naming objects by matching them with a vocabulary of known objects). On the other hand, the fact that modern VNNs consistently label an elephant with white fur as a polar bear and fail to recognize a cow on a beach as a cow suggests that they have managed to solve Papert's final goal while ignoring the first two goals: figure ground segmentation and region description.

The relatively low-level nature of the classification strategy used by VNNs should not be viewed as a criticism of VNNs: They automatically extract certain low-level cues that solve classification tasks at a high level (even for unseen images). This is merely an example of shortcut learning (Geirhos et al. 2020), which can occur in any system that learns automatically from examples, including humans. For example, deep neural networks trained to look for logical contradictions in texts can attain high performance by simply checking for the presence or absence of the word "not" (Dogra et al. 2024). The blame for such shortcuts should not be placed on the neural networks, but rather on the biases in the datasets used to train and evaluate them.

One commonly applied method to increase the robustness of VNNs is to use extensive data augmentation and try to expose the VNNs during training to examples of perturbations for which we want them to be robust. In the context of adversarial examples, this is known as adversarial training (Madry et al. 2018), a procedure whereby adversarial examples are computed during training and added to the training set. Although this method can increase robustness to the same type of perturbation seen during training (e.g., Gaziv et al. 2024), it significantly slows down training, which may explain why none of the VNNs in **Table 1** use adversarial training. More significantly, it has been reported that "most adversarial training methods can only defend against a specific attack that they are trained with" (Nie et al. 2022, p. 16805): Even after adversarial training, it is easy to fool the VNN with an attack that is different from what it saw during training. A similar effect has been reported for other attempts to increase robustness by data augmentation: Robustness is indeed increased to the particular perturbation for which we train on but not to other perturbations (Rebuffi et al. 2021, Gunasekar 2022).

Given that learning from examples is susceptible to shortcut learning and that all successful VNNs are constructed almost exclusively by learning from examples, will VNNs ever learn to perform classification robustly? The predominant view in today's computer vision community is that robustness will be achieved by using larger and more varied datasets (Radford et al. 2021). While it is hard to speculate about the future, the results we have reviewed seem inconsistent with this view. Consider, for example, the sensitivity to small pixel shifts shown in **Table 3**. First, the vanilla ViT, trained with hundreds of millions of examples, is more sensitive to single-pixel shifts than ResNet50, which was trained on a million examples. Second, even models that were trained on billions of examples and use data augmentation did not learn to be robust to single-pixel

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translations of the image. This does not rule out the possibility that robustness will emerge with even more images, but, in our opinion, it makes it unlikely.

SUMMARY POINTS

1. Visual neural networks (VNNs) have learned relatively low-level cues that are extremely effective at classifying internet images but are ineffective at classifying many other images that humans can classify effortlessly.
2. VNNs are often fooled by image changes to which human object recognition is largely invariant (e.g., the change of a few pixels in the image or a change of the background or illumination).
3. VNNs can be invariant to very large image manipulations that totally disrupt human performance (e.g., randomly permuting the patches of an image).

FUTURE ISSUES

1. The predominant view in today's computer vision community is that robustness will be achieved by using larger models and more varied datasets.
2. In our opinion, simply scaling up the size of the datasets and models is not sufficient for obtaining human-like robust visual performance. Rather, robustness is more likely to be achieved with a combination of deep learning and classical approaches based on understanding image formation and mathematical formulations.

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